Sage: Smart Academic Hiring System

Literature Review Report

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Date: October 2025

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We declare that this project titled "SAGE : Smart Academic Hiring System", submitted as a requirement for the award of the degree of Bachelor of Science in Computer Science, does not contain any material previously submitted for a degree in any university; and that to the best of our knowledge, it does not contain any materials previously published or written by another person except where due reference is made in the text.

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Abstract

This document presents the literature review for the **Sage (Smart Academic Hiring System)** Final Year Project. Sage automates faculty hiring at FAST University using resume parsing, candidate-job matching via an improved Gale–Shapley algorithm, automated LLM-based testing, and AI interview evaluation focusing on non-technical attributes such as confidence and personality. This review examines related research on resume parsing, candidate-job matching, interview automation, and fairness in AI recruitment, highlighting existing gaps addressed by Sage.

Contents

Chapter 1

Introduction

1.1 Background

Recruitment and selection of faculty is a time-consuming and often subjective process. Universities typically rely on manual screening of CVs, ad-hoc interviews, and inconsistent evaluation criteria. Recent advances in Natural Language Processing (NLP), Machine Learning (ML), and Large Language Models (LLMs) now enable automation of hiring tasks such as CV parsing, semantic matching between candidate profiles and job descriptions, automated testing, and even preliminary AI-conducted interviews.

1.2 Motivation

Sage automates the entire recruitment process for academic hiring at FAST University. By integrating structured CV parsing, a stable-matching algorithm (Gale–Shapley), LLM-powered test generation and grading, and an AI interviewer for behavioral assessment, Sage reduces HR workload, improves fairness, and ensures consistency in faculty selection.

1.3 Scope of the Literature Review

This literature review focuses on:

- Resume/CV parsing and NLP-based information extraction.
- Candidate–job matching via the Gale–Shapley stable matching algorithm.
- Automated testing using LLMs for question generation and grading.
- AI-driven interview evaluation.
- Fairness and bias mitigation in automated hiring.

Chapter 2

Review of Literature

2.1 Method for Selecting Papers

Peer-reviewed studies from IEEE, Springer, Elsevier, and arXiv (2020–2025) were reviewed. Selection criteria included technical relevance to AI recruitment, matching algorithms, and ethical automation.

2.2 Comparative Analysis of Related Work

Table 2.1 summarizes major studies relevant to Sage—covering Gale–Shapley algorithm variants, resume parsing systems, AI interview bots, and fairness in automated recruitment.

Year	Paper / System	Methodology	Dataset / Evaluation	Main Findings	Gaps / Limitations
2020 (Est.)	**Performance-based Stable Matching using Gale–Shapley Algorithm**	Application of the Gale–Shapley algorithm for employee placement using performance metrics and institutional preferences.	Conceptual study for employee position balancing.	Demonstrated stable matching reduces dissatisfaction from misplacement.	Focused on internal placement; does not address external hiring like **Sage**.
2021	**Interview Bot for Improving Human Resource Management**	AI/ML-driven chatbot automating interview conduction and evaluation.	Conceptual design and feasibility study.	Proved AI chatbots can reduce bias and improve interview scalability.	Lacked large-scale testing; text-based only, not multimodal like **Sage**.
2022	**Enhancement of Gale–Shapley Algorithm with Imbalanced Sets**	Modified Gale–Shapley for unbalanced sets (candidates vs jobs) using duplication for stable pairing.	Theoretical study.	Extended stable matching to practical hiring imbalance.	Relied on duplication; **Sage** instead uses weighted scoring to balance matches.
2022	**People versus Machines: Introducing the HIRE Framework**	Systematic review of 22 empirical AI recruitment studies proposing HIRE framework.	Literature synthesis.	Found AI comparable or superior to human recruiters in efficiency.	Identified mistrust and ethical concerns; conceptual framework only.
2024	**Automated Resume Parsing and Ranking using NLP**	Transformer (BERT)-based NLP model for resume parsing and ranking.	Internet resume datasets.	Improved accuracy for skill and education extraction.	General-purpose parser; not optimized for academic CVs as in **Sage**.
2024	**Putting Gale & Shapley to Work: Guaranteeing Stability Through Learning**	Learning-based stable matching ensuring stability under uncertain preferences.	Theoretical simulations.	Provided stability guarantees in uncertain preference settings.	Purely theoretical; lacks domain weighting like in **Sage**.
2025	**Optimizing Recruitment with AI: A Smarter Approach to Candidate Evaluation**	AI-driven recruitment pipeline automating screening and interviews.	System design prototype.	Increased efficiency and accuracy in recruitment.	General recruitment; lacks academic-domain adaptation of **Sage**.
2025	**Fairness in AI-Driven Recruitment: Challenges, Metrics, Methods, and Future Directions**	Systematic review of bias metrics and mitigation in AI hiring.	Review of 120+ studies.	Summarized fairness metrics and mitigation frameworks.	Conceptual; **Sage** incorporates fairness-aware scoring to address this.

Table 2.1: Comparative Literature Review for **Sage – Smart Academic Hiring System**.

Chapter 3

Project Vision

3.1 Problem Statement

Manual shortlisting and interview scheduling in academic hiring is slow and inconsistent. Candidate evaluation is often subjective and hard to audit. Sage addresses these issues by automating extraction, matching, testing, and preliminary interviewing while keeping HR oversight.

3.2 Business Opportunity

Automation reduces HR time, speeds up hiring cycles, and yields more consistent candidate assessments. For a large university with recurring hiring cycles, Sage can materially reduce administrative costs and improve candidate–role fit.

3.3 Objectives

1. Build a CV parsing module for academic resumes.
2. Implement a candidate-job matching ranking engine.
3. Integrate LLM-based test generation and auto-grading for objective/short-answer questions.
4. Develop an AI interviewer to evaluate non-technical attributes and generate recorded sessions.
5. Provide transparent dashboards for HR with audit trails and candidate explanations.

3.4 Project Scope

Sage focuses on initial shortlisting, automated testing, AI interviewing for non-technical attributes, and final rank-generation for HR review. It does not perform final hiring approvals or legally binding decisions.

3.5 Constraints

Data privacy regulations, limited access to diverse labeled interview datasets, variability in CV formats, and the need to avoid biased models are the main constraints.

3.6 Stakeholders

- HR Staff — primary users who control hiring workflows.
- Candidates — supply CVs and take tests/interviews.
- Department Heads — view ranked lists and interview recordings.
- University Administration — final approvals and policy oversight.

Chapter 4

Diagrams and Placeholders

The following pages are placeholders for diagrams that you will insert later. Replace the framed boxes with actual diagrams (PDF/PNG/SVG) and adjust captions.

4.1 Use Case Diagram

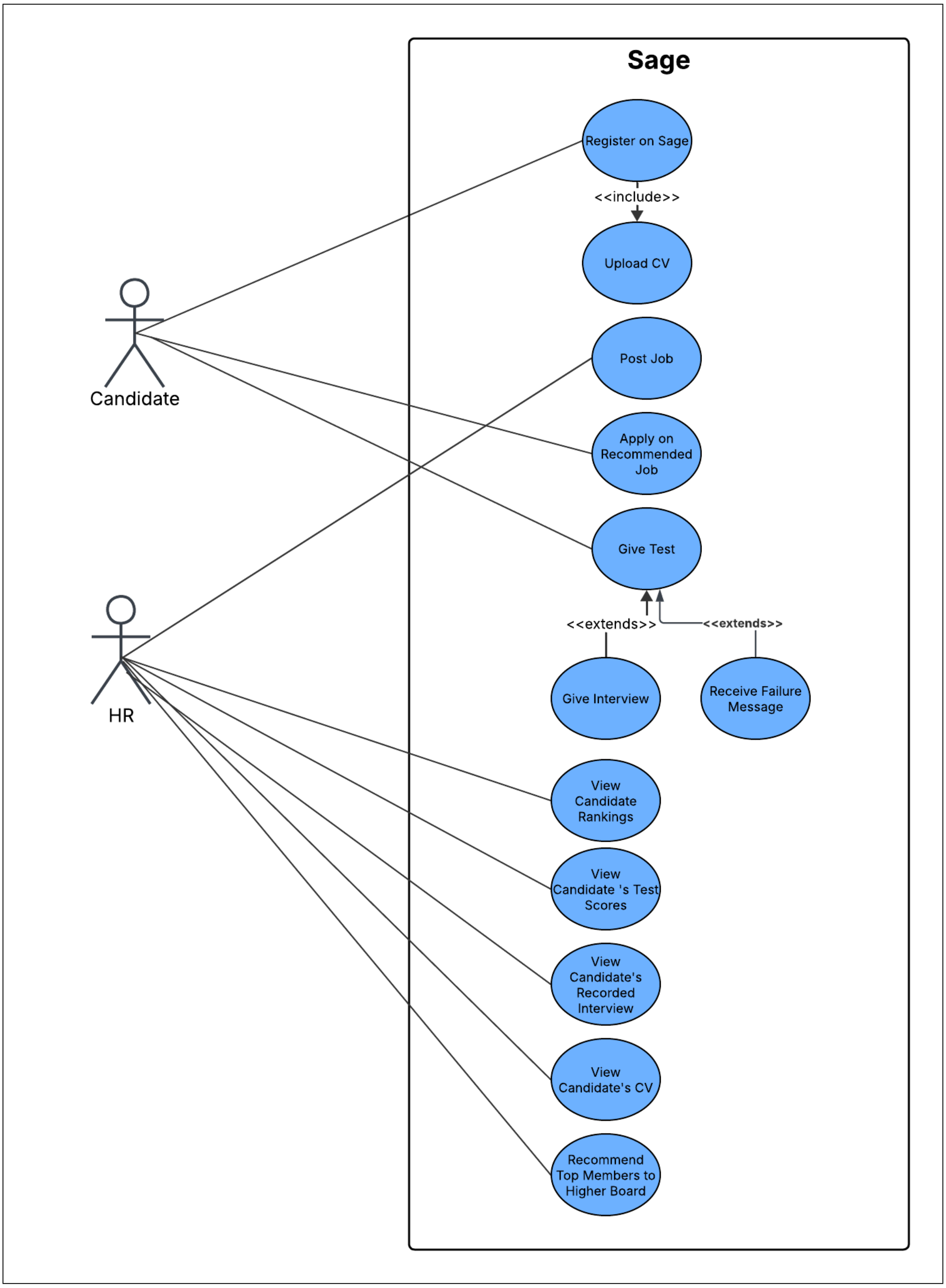

Figure 4.1: Use Case Diagram: High-Level System Interactions.

4.2 Sequence Diagram

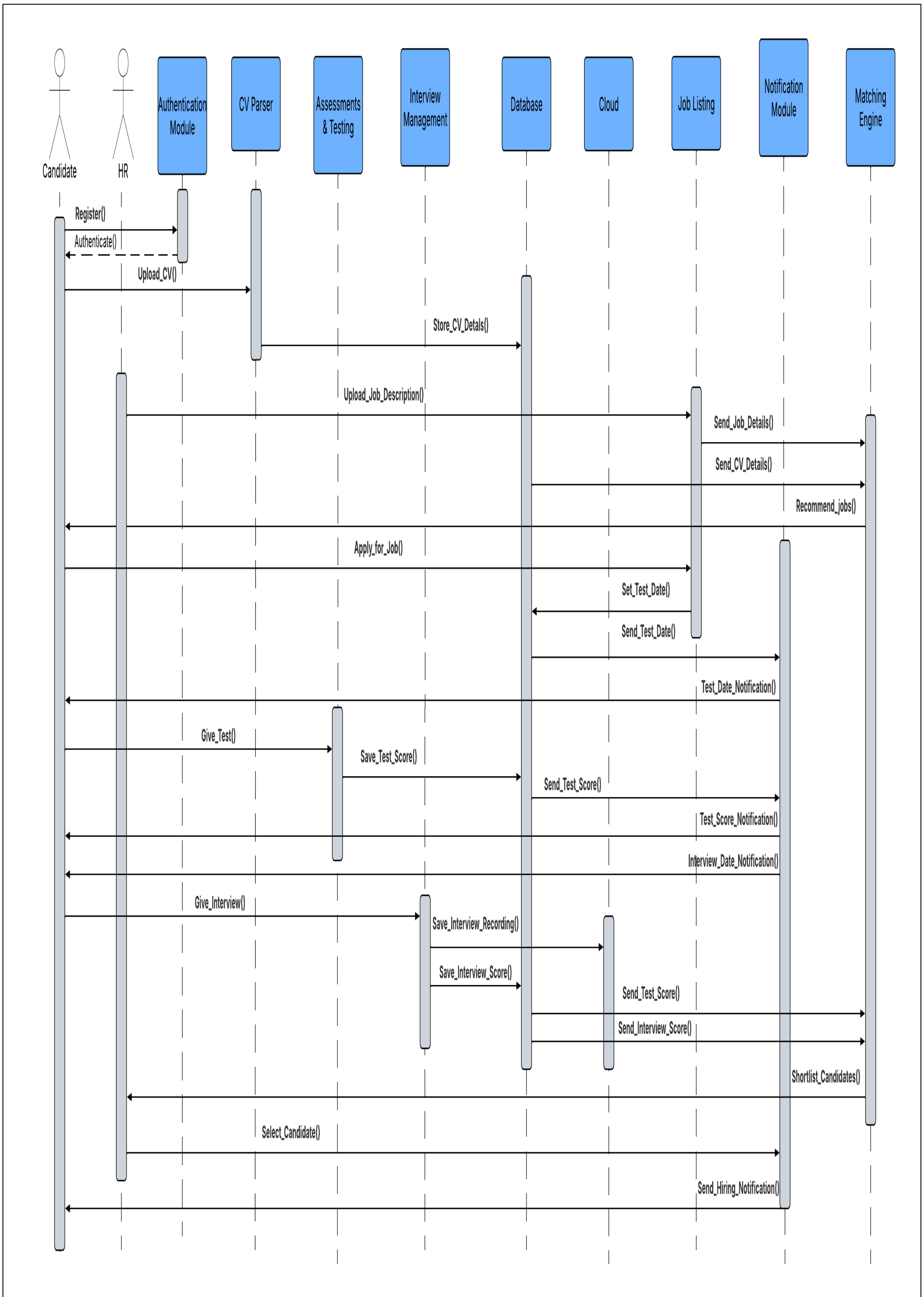

Figure 4.2: Sequence Diagram: Candidate Registration and Screening Process.

4.3 Data Flow Diagram (DFD)

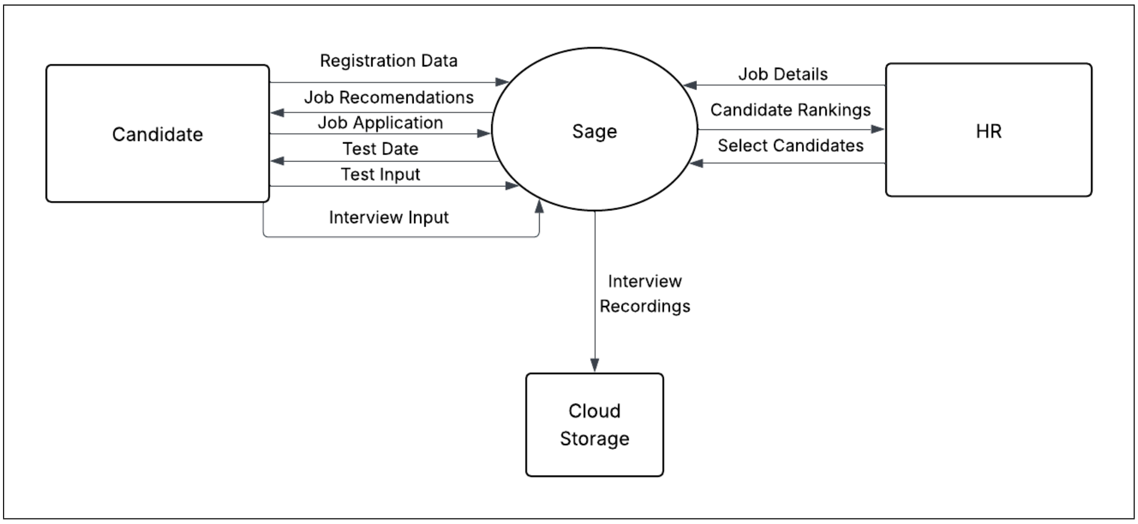

Figure 4.3: Data Flow Diagram — Level 1: System Data Flow Overview.

4.4 Activity Diagrams

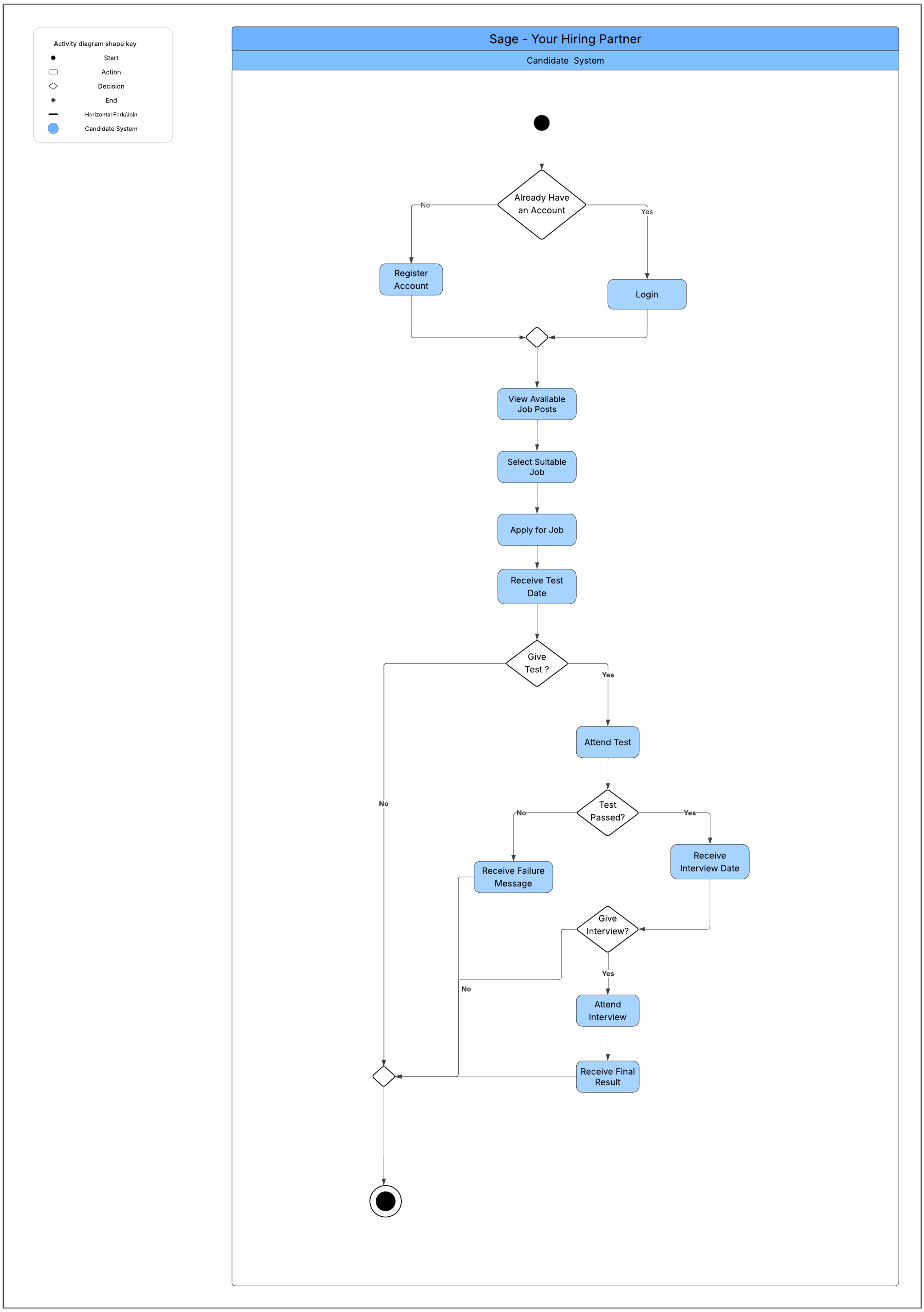

Figure 4.4: Activity Diagram: Candidate-System Interaction for Application.

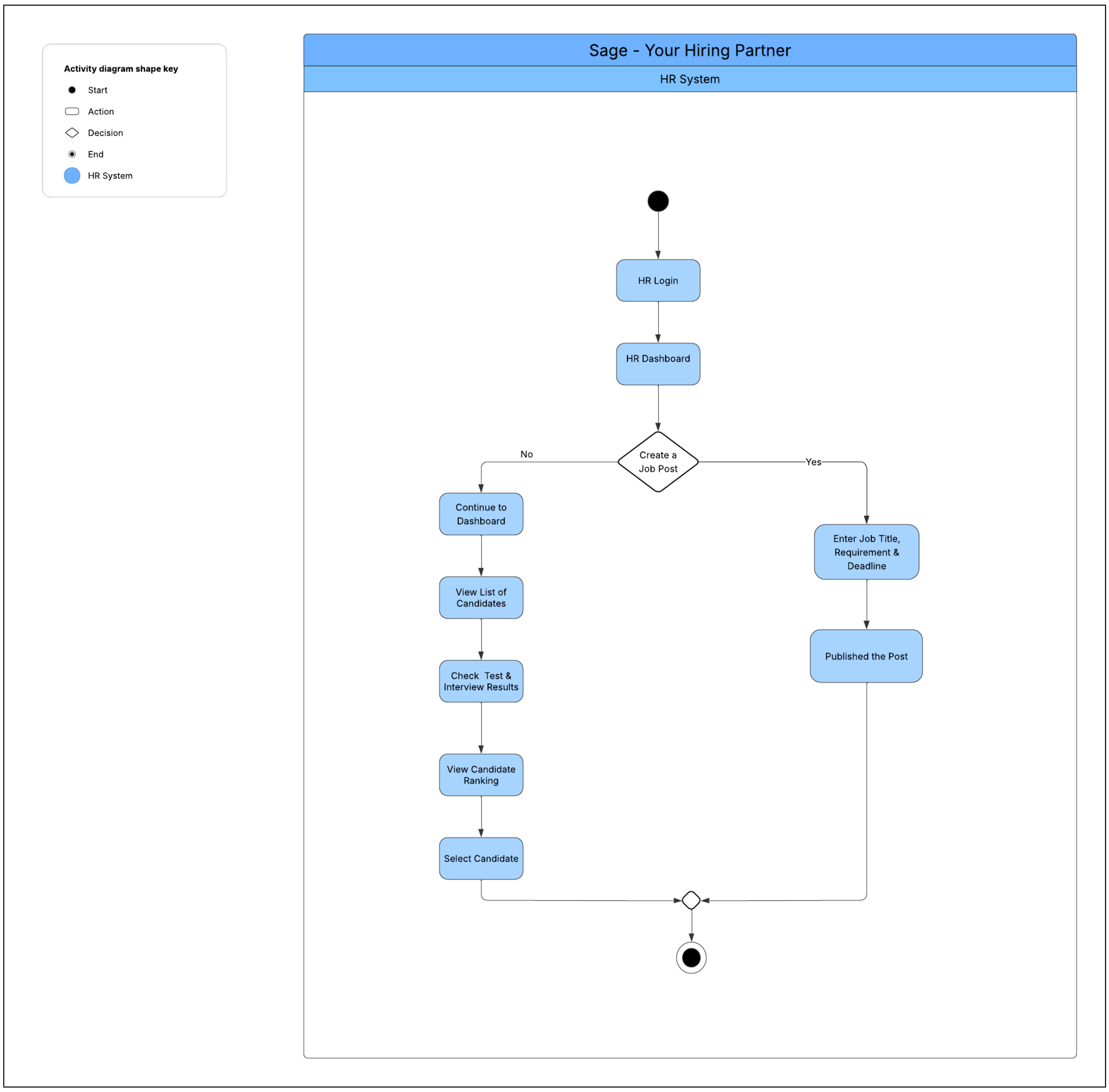

Figure 4.5: Activity Diagram: HR Staff Review and Final Matching Process.

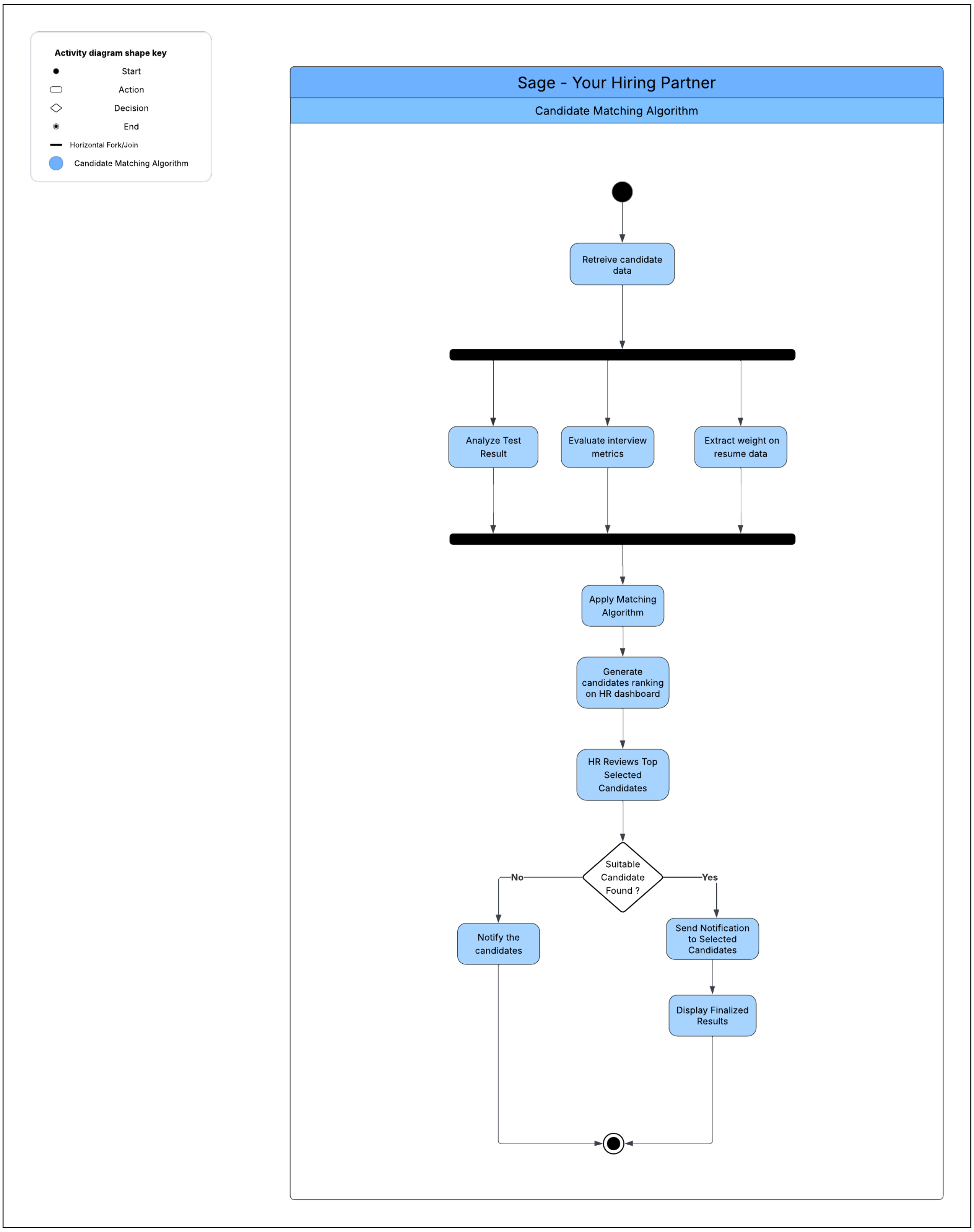

Figure 4.6: Activity Diagram: Candidate Interview Workflow.

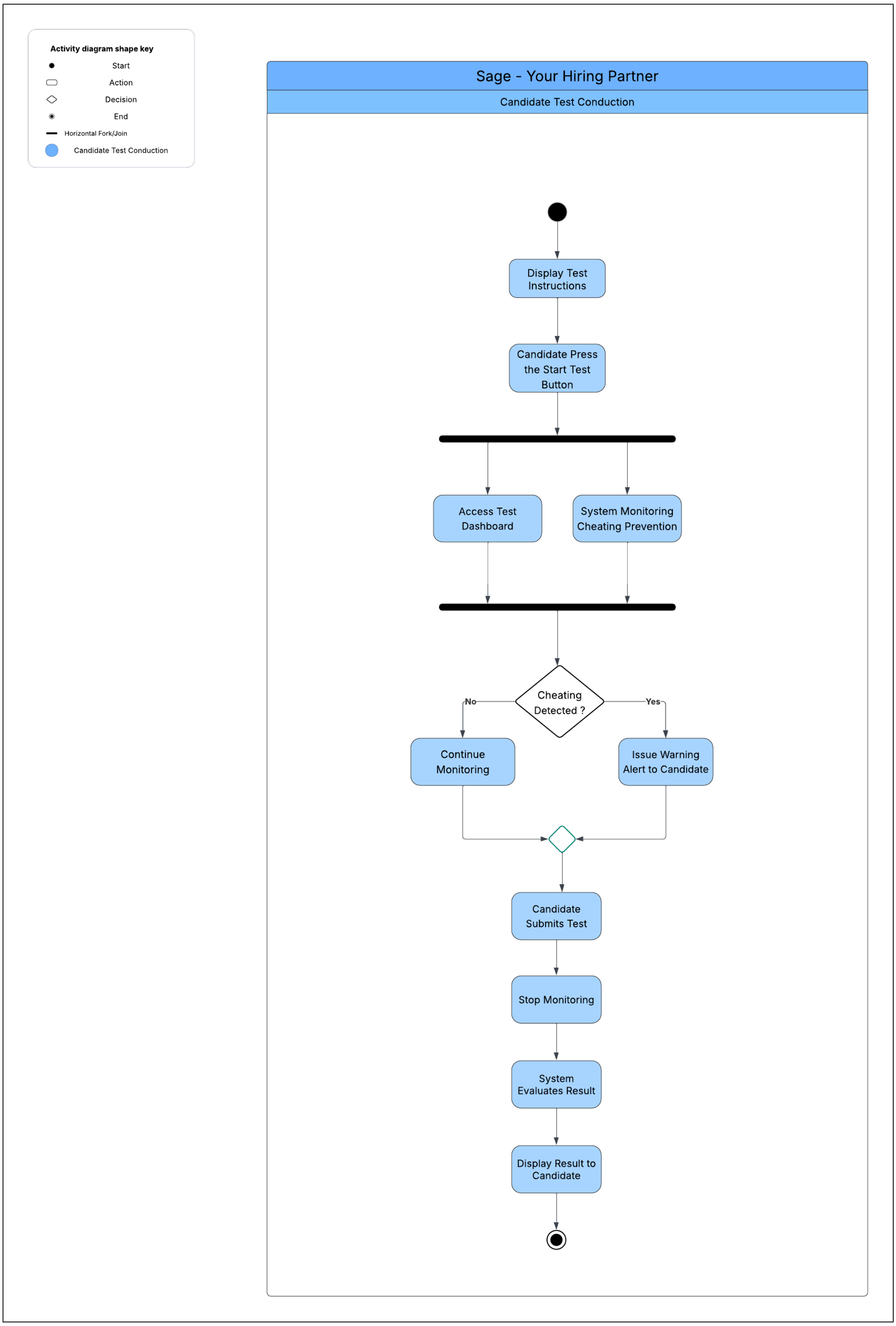

Figure 4.7: Activity Diagram: Candidate Test Generation and Submission.

4.5 Layer / Architecture Diagram

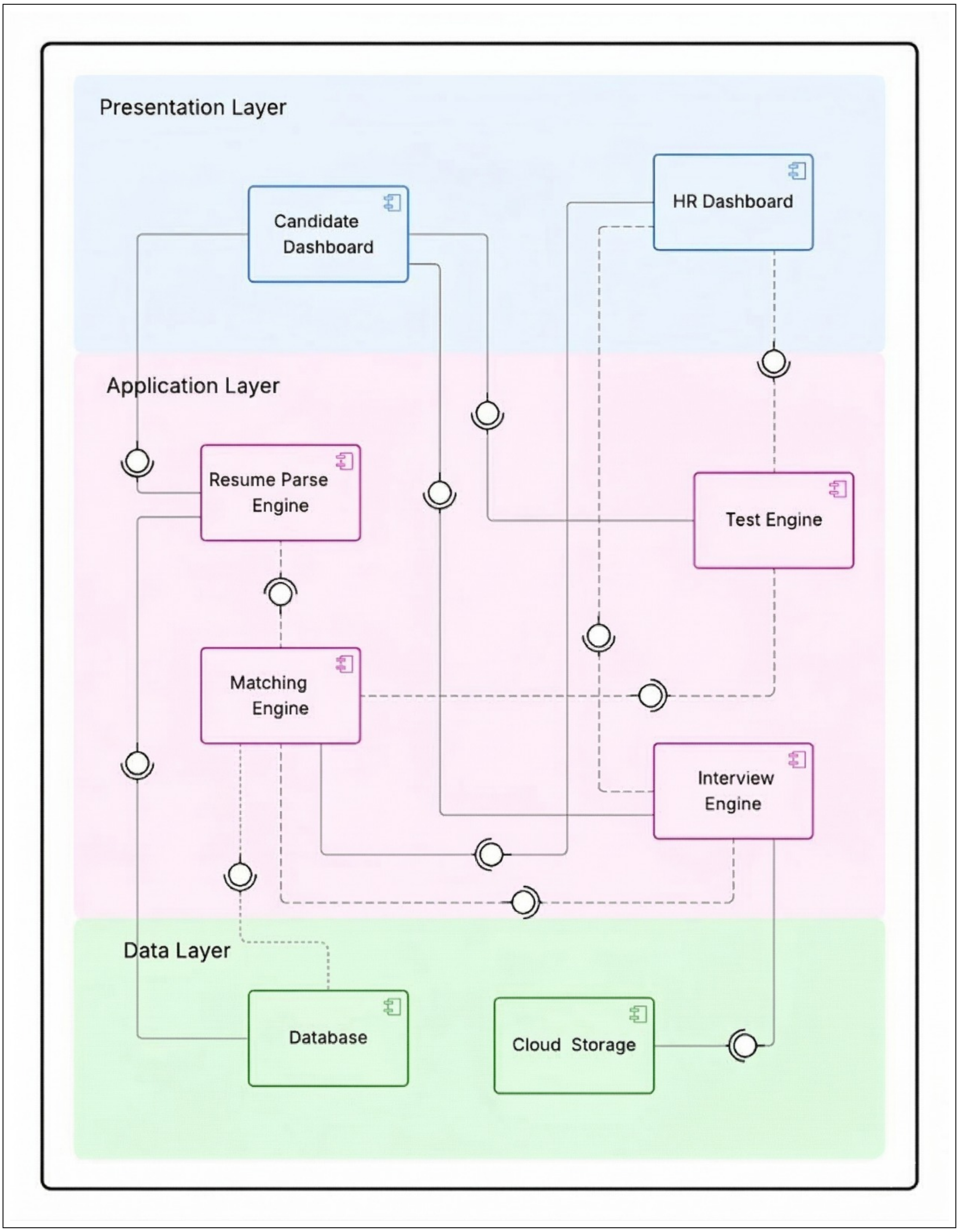

Figure 4.8: Activity Diagram: Candidate Test Generation and Submission.

Chapter 5

Methodological Notes for Implementation

5.1 CV Parsing

Suggested approach:

- Preprocessing using regex.
- Heuristic post-processing to unify degree names, map skill synonyms.
- Optional: fine-tune Layout-aware models on a subset of academic CVs.

5.2 Matching and Ranking

- Use sentence / document embeddings for semantic similarity between job description and candidate profile.
- Combine similarity score with weighted heuristics: university ranking, years of experience, publication count, teaching experience, domain certifications.
- Train a learning-to-rank model (LambdaMART or pairwise ranker) if labeled match data becomes available.

5.3 Automated Testing

- Use existing question bank. Generate variations via LLM prompt templates.
- For objective questions: auto-grade.
- Set threshold (e.g., 40%) for progression to interview stage (configurable).

5.4 AI Interviewer

- Design interview flows focusing on behavioral prompts and situational questions.

- Capture audio and transcripts; apply multimodal analysis for confidence (prosodic features) and text-based personality signals.
- Always record interviews and allow HR to review and override AI assessments.

5.5 Fairness and Explainability

- Log model decisions and feature contributions for each candidate (SHAP or local explanations).
- Implement fairness audits (statistical parity, disparate impact) on historical data.
- Provide configurable attribute weights to allow policy-driven adjustments.

Chapter 6

Iteration Plan

This chapter outlines the planned development approach for the SAGE (Smart Academic Hiring System) project. The work is divided into four major checkpoints across FYP–1 and FYP–2. Each phase has specific goals related to system analysis, design, implementation, and evaluation.

6.1 Midterm FYP 1

During this phase, the focus is on understanding the academic hiring domain, studying existing AI recruitment systems, collecting requirements from HR stakeholders, and completing the literature review. Initial system design artifacts—such as use case diagrams, system flow, and high-level architecture—are also produced to build a foundational understanding of SAGE.

6.2 Final FYP 1

This phase includes the finalization of the Software Requirements Specification (SRS) and the development of core system components such as the CV parsing module, candidate–job profiling structure, and early matching prototype. The UI layout for HR and candidates is also drafted, and basic backend endpoints are implemented.

6.3 Midterm FYP 2

At this stage, major system functionality is developed. This includes implementing the improved Gale–Shapley matching engine, integrating LLM-based screening test generation, building the AI interview module, and connecting parsed CV data with ranking logic. Initial internal testing of all modules is conducted to validate correctness.

6.4 Final FYP 2

The final phase includes polishing the system, completing integration of all modules, performing system-wide testing, preparing the user manual, and deploying the SAGE platform for demonstration. Performance optimization and security verification in the backend are also completed.

Chapter 7

User Interface Design and Test Cases

This chapter presents the graphical user interfaces of the Smart Academic Hiring System. The system consists of two primary interfaces: the **Candidate Frontend Application** and the **HR Dashboard**. The candidate application allows job seekers to upload resumes, manage profiles, browse job listings, apply for positions, and take automated assessments. The HR dashboard provides recruiters with tools to monitor applicants, manage job postings, analyze hiring data, and control the recruitment pipeline.

The interfaces are designed with usability and simplicity in mind so that both candidates and HR staff can interact with the system efficiently.

7.1 Candidate Frontend Interface

The candidate interface is a web-based application developed using **React, TypeScript, and Vite**. It allows candidates to interact with the hiring system, manage their personal profiles, and participate in the recruitment process.

The main features of the candidate interface include:

- Resume upload and automatic profile generation
- Job browsing and application submission
- Application tracking
- Online Test
- Profile management

7.1.1 Main Landing Page

The landing page serves as the main entry point of the platform. It introduces users to the Smart Academic Hiring System and provides an overview of the platform's purpose and functionality. From this page, users can navigate to candidate registration, login, or explore available job opportunities.

The landing page typically highlights important information such as platform features, benefits for candidates, and quick navigation options to start the application process.

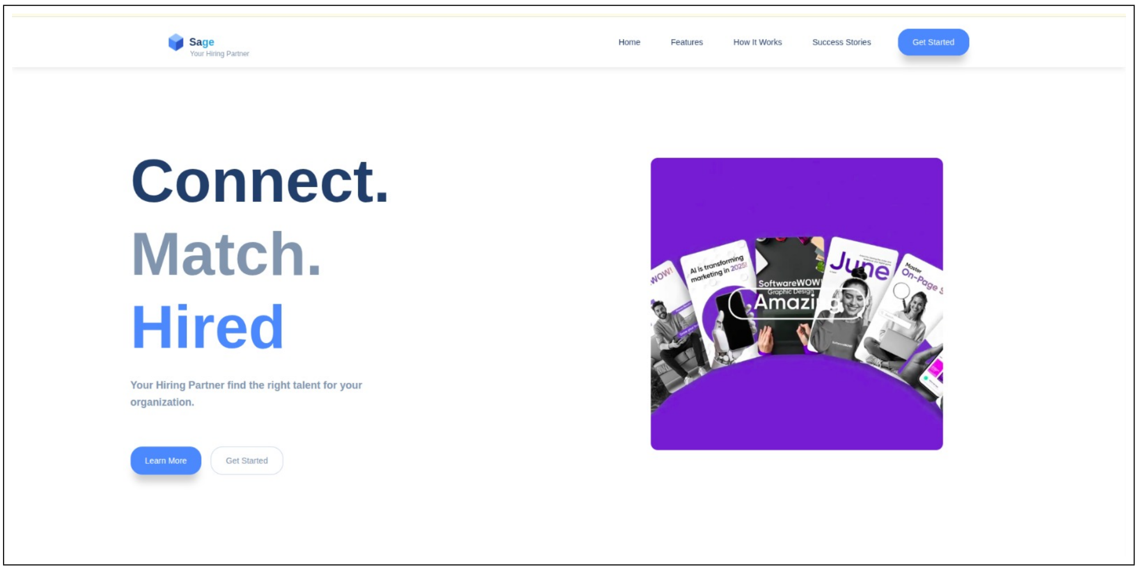

Figure 7.1: Main Landing Page of the Smart Academic Hiring System

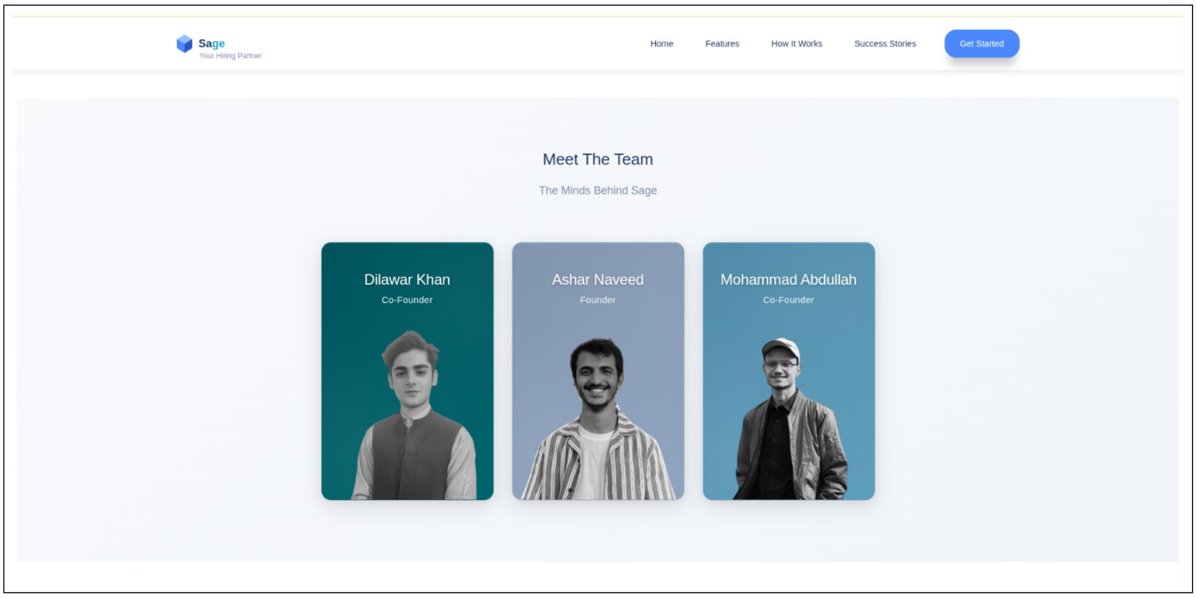

Figure 7.2: Main Landing Page of the Smart Academic Hiring System

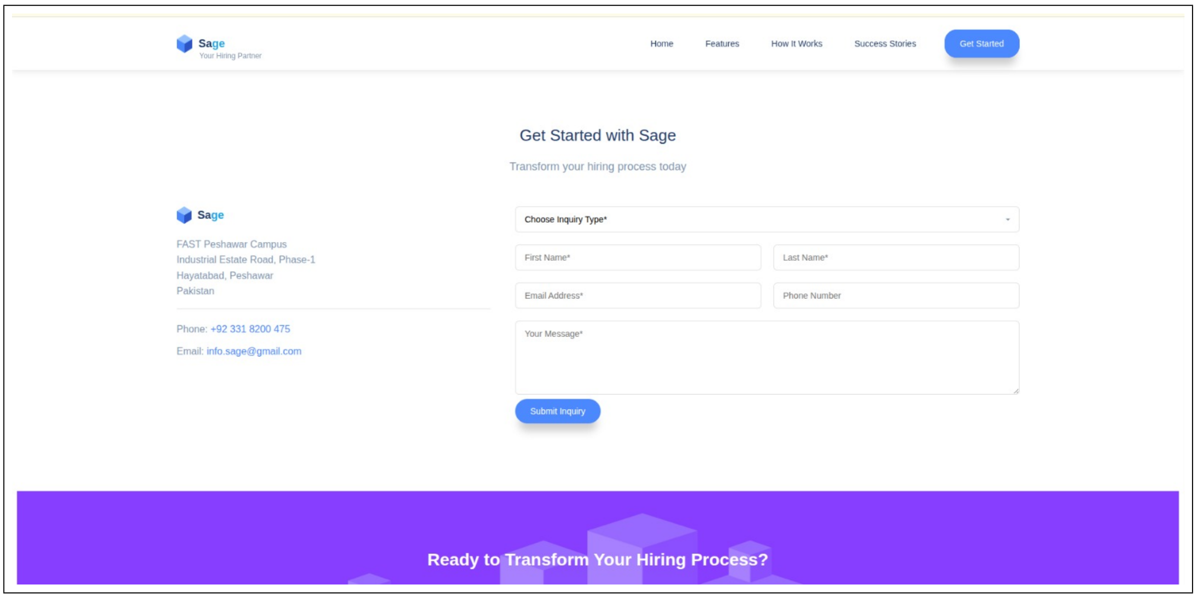

Figure 7.3: Main Landing Page of the Smart Academic Hiring System

7.1.2 Candidate Login Page

The login page allows candidates to authenticate themselves before accessing the platform. After successful login, the user is redirected to their dashboard where they can manage their applications and profile.

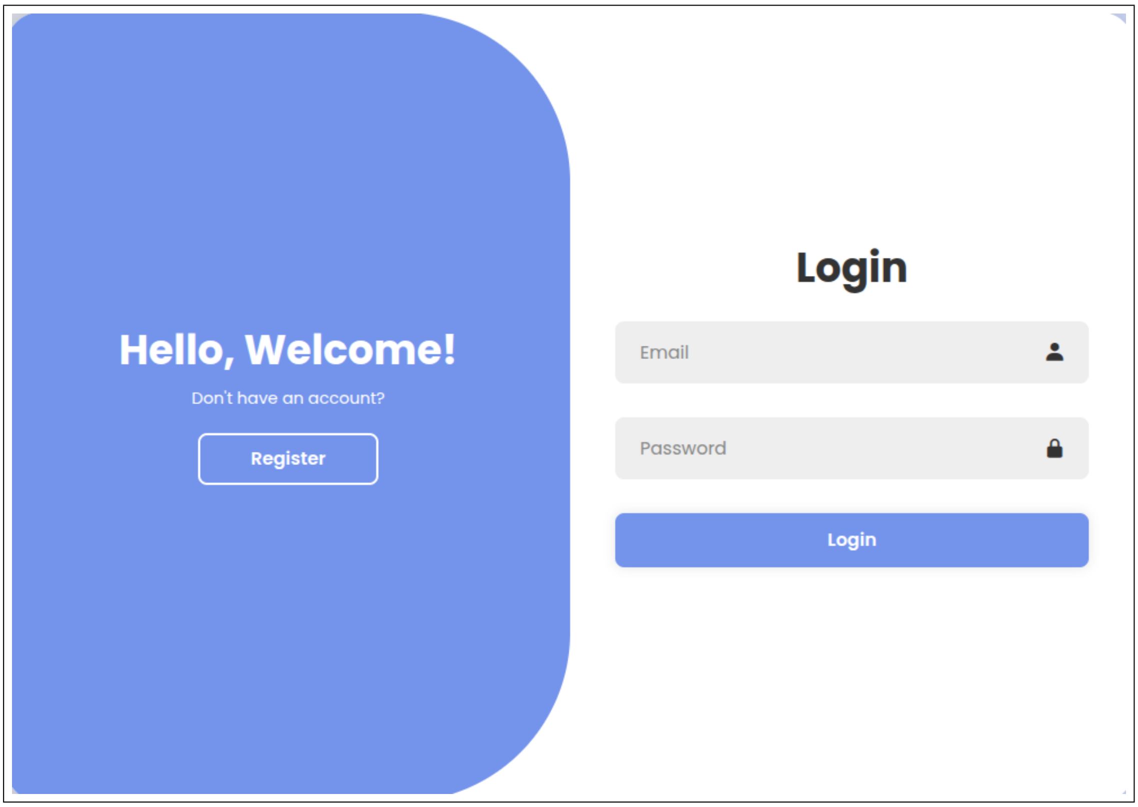

Figure 7.4: Main Landing Page of the Smart Academic Hiring System

7.1.3 Resume Upload and Profile Creation

Candidates can upload their CV in PDF format. The system parses the uploaded resume and automatically extracts information such as education, skills, work experience, and projects. This information is used to populate the candidate's profile.

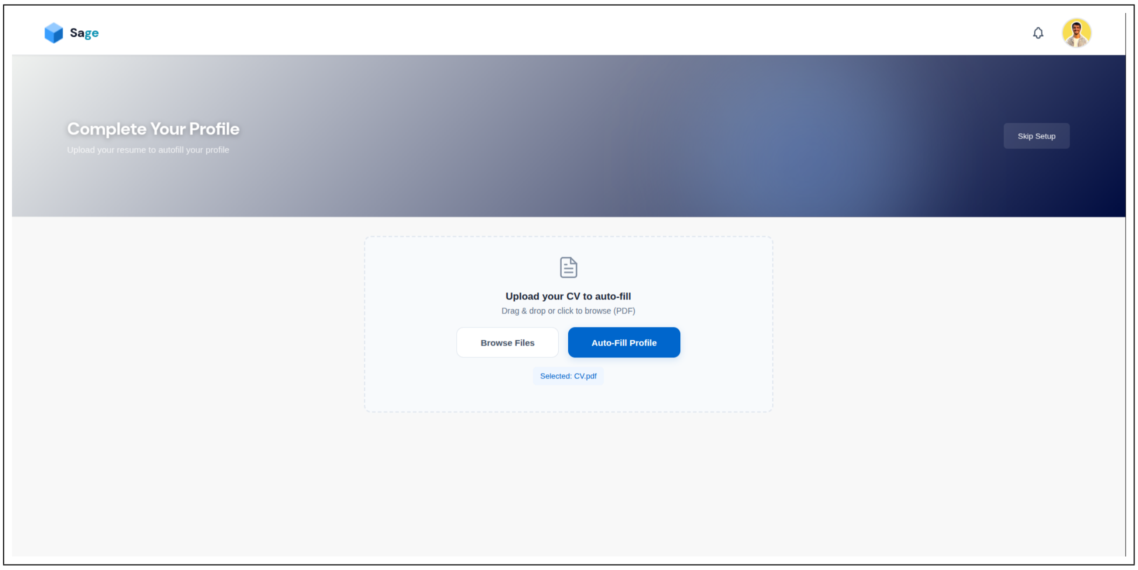

Figure 7.5: Main Landing Page of the Smart Academic Hiring System

7.1.4 Job Listings Page

The job listings page displays all available job postings. Candidates can browse jobs, view detailed descriptions, and apply directly from the interface.

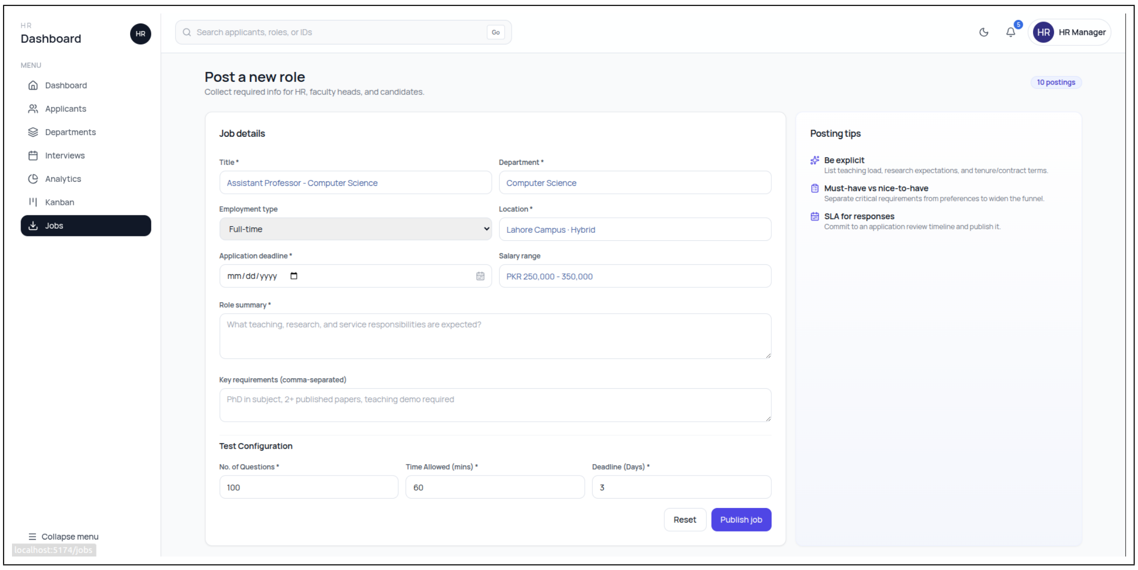

Figure 7.6: Main Landing Page of the Smart Academic Hiring System

7.1.5 Applications Tracking Page

This page allows candidates to monitor the status of their submitted applications. The system displays different stages such as applied, under review, interview scheduled, and final decision.

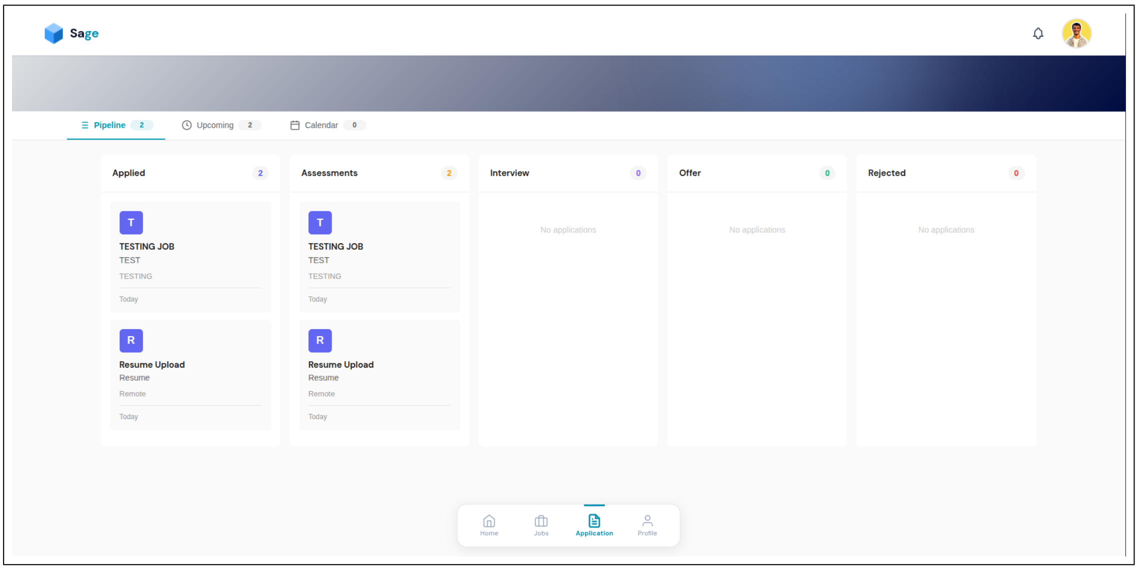

Figure 7.7: Main Landing Page of the Smart Academic Hiring System

7.2 HR Dashboard Interface

The HR dashboard is designed for recruiters and administrators to manage the recruitment process. It provides tools for monitoring applicants, managing job postings, viewing analytics, and controlling the hiring pipeline.

The HR dashboard is developed using **React and Vite** and communicates with the backend through REST APIs.

Key features of the HR dashboard include:

- HR login and authentication
- Applicant management and filtering
- Job posting creation and management
- Interview scheduling and recordings
- Visual hiring pipeline (Kanban board)
- Recruitment analytics dashboard

7.2.1 HR Dashboard Overview

The main dashboard provides HR personnel with an overview of recruitment statistics, including the number of applicants, active job postings, and recent hiring activities.

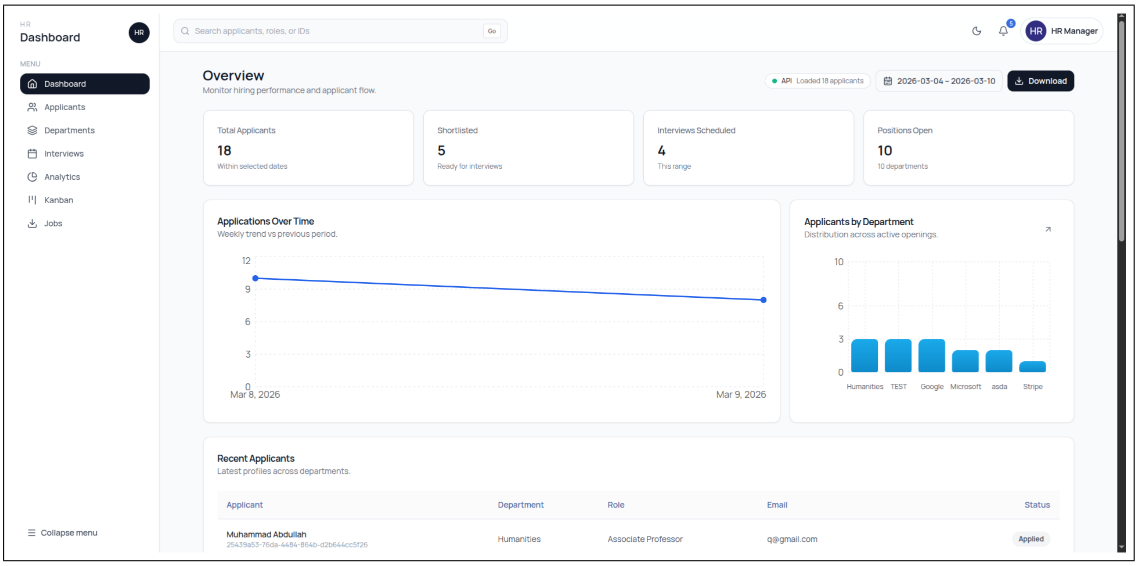

Figure 7.8: Main Landing Page of the Smart Academic Hiring System

7.2.2 Applicants Management Page

The applicants page displays a list of all candidates who have applied for jobs. HR staff can filter applicants based on role, department, or application status. Detailed candidate profiles including resumes and test scores can be viewed from this interface.

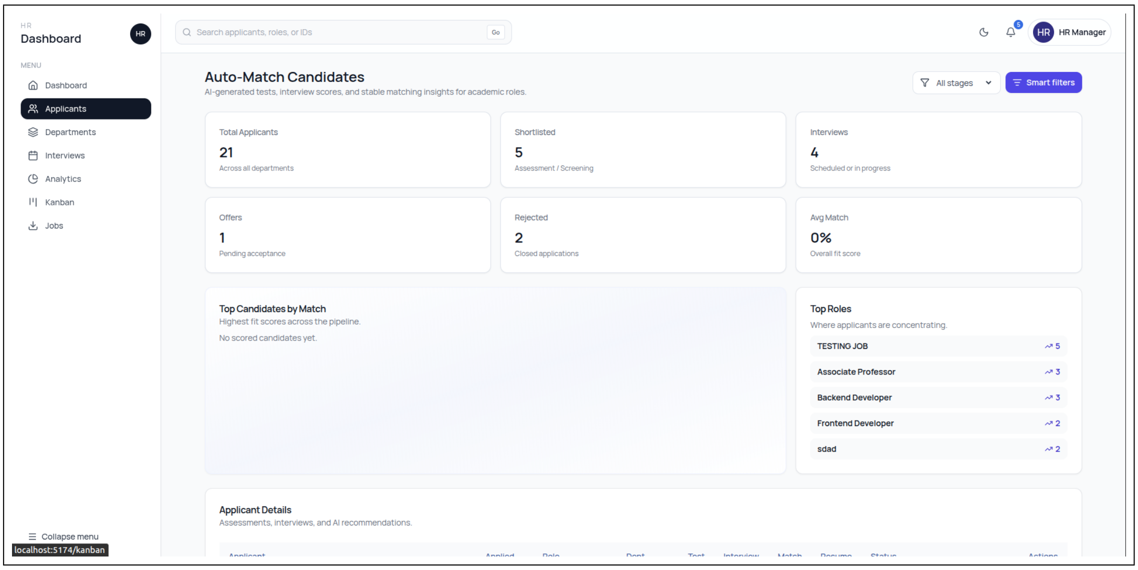

Figure 7.9: Main Landing Page of the Smart Academic Hiring System

7.2.3 Job Management Page

The job management interface allows HR to create new job postings, update job descriptions, and delete outdated listings. Each job posting can also display the applicants who have applied for that role.

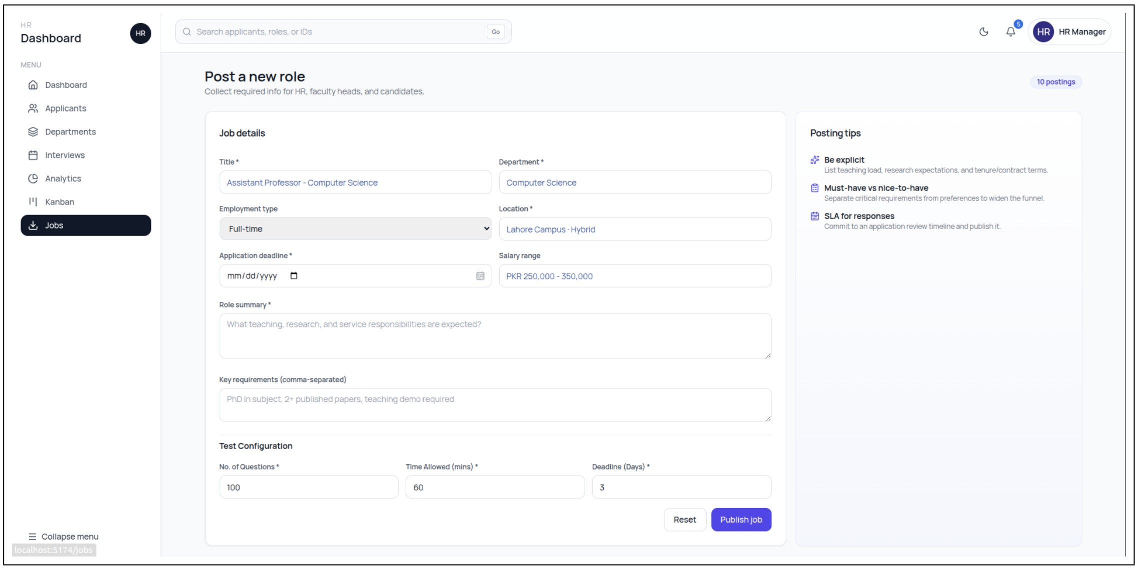

Figure 7.10: Main Landing Page of the Smart Academic Hiring System

7.2.4 Hiring Pipeline (Kanban Board)

The Kanban board visually represents the recruitment pipeline. Candidates move through different stages such as application review, interview, and final selection. HR staff can update applicant status using simple drag-and-drop or action buttons.

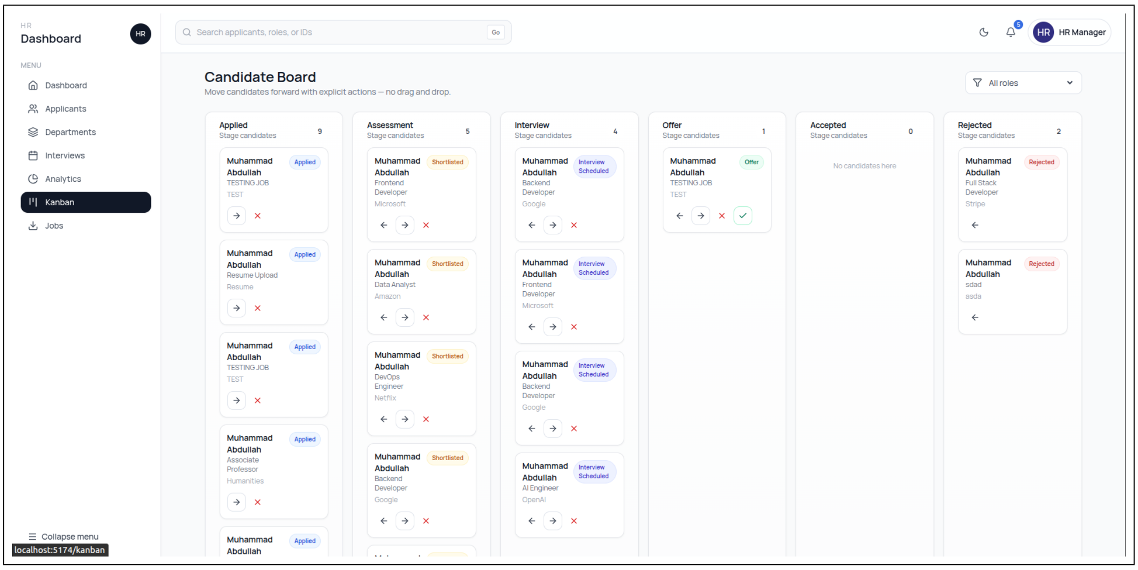

Figure 7.11: Main Landing Page of the Smart Academic Hiring System

7.3 System Test Cases

To ensure the reliability and correctness of the Smart Academic Hiring System, several functional and integration test cases were designed and executed across different modules of the system. These tests validate the behavior of the CV parsing module, the candidate–job matching engine, the RAG-based test generation module, and the interaction between the frontend and backend components.

CV Parsing Module Tests

Test Case CV-001: Basic Information Extraction

This test verifies that the CV parser correctly extracts essential contact information from a standard resume layout.

Expected Outcome: The system correctly identifies and stores the candidate's Name, Email, and Phone number.

Test Case CV-002: Skill Normalization

This test checks whether the parser normalizes different textual expressions referring to the same skill.

Expected Outcome: Phrases such as "Python Developer" and "Proficient in Python" are mapped to the standardized skill *Python*.

Test Case CV-003: Complex Resume Formatting

This test evaluates the parser's ability to process resumes containing multi-column layouts or tabular sections.

Expected Outcome: The parser correctly extracts information without merging unrelated text or breaking section structure.

Test Case CV-004: Section Identification

This test verifies that the parser correctly separates logical sections within a CV.

Expected Outcome: Work Experience and Projects sections are detected and stored separately.

Candidate–Job Matching Engine Tests

Test Case MATCH-001: Technology Synonym Matching

This test validates that the matching engine understands technology relationships and synonyms.

Expected Outcome: A candidate with experience in Django is matched with jobs requiring Web Framework expertise.

Test Case MATCH-003: Similarity Threshold Filtering

This test ensures that irrelevant job recommendations are filtered out.

Expected Outcome: Jobs with semantic similarity lower than 20 are not recommended to the candidate.

Test Case MATCH-004: Direct Skill Overlap Bonus

This test verifies that exact skill matches increase the candidate ranking score.

Expected Outcome: Exact matches such as React receive an additional 50

RAG-Based Test Generation Module Tests

Test Case TEST-001: LLM Question Generation

This test evaluates whether the Retrieval-Augmented Generation (RAG) system produces properly formatted multiple-choice questions.

Expected Outcome: The system generates valid JSON containing one question, four answer options, and a single correct answer key.

Test Case TEST-002: Fallback Question Retrieval

This test checks the fallback mechanism when the LLM service is unavailable.

Expected Outcome: The system retrieves questions from the internal JSON question bank.

Test Case TEST-003: Secure Backend Scoring

This test ensures that test scores cannot be manipulated from the frontend.

Expected Outcome: The backend recalculates the score using submitted answers and ignores any score values sent from the frontend.

Test Case TEST-004: Test Status Transition

This test validates the status transition rules applied after test completion.

Expected Outcome: Candidates with scores greater than or equal to 40

Integration and User Interface Tests

Test Case UI-001: Candidate Onboarding Flow

This test verifies the onboarding workflow for new candidates.

Expected Outcome: After uploading a CV, the candidate is redirected to the profile page where fields are automatically filled using parsed resume data.

Test Case UI-002: Test Timer Synchronization

This test ensures synchronization between the frontend test timer and the backend session control.

Expected Outcome: The test automatically submits when the timer expires even if the candidate does not manually submit it.

Test Case UI-003: HR Dashboard Ranking View

This test validates the ranking functionality in the HR dashboard.

Expected Outcome: HR users can view applicants sorted according to their calculated relevance score.

7.4 Summary

The user interfaces of the Smart Academic Hiring System are designed to simplify the recruitment process for both candidates and HR staff. The candidate application focuses on ease of job search and application management, while the HR dashboard provides powerful tools for managing applicants, job postings, and hiring workflows. Together, these interfaces provide a complete and efficient recruitment platform.

Chapter 8

Conclusion

This review shows that while components for automated hiring exist, an integrated, auditable system tailored to academic hiring (with CV heterogeneity, publications, and university-quality signals) is still missing. Sage plans to combine best practices from parsing, semantic matching, LLM-based assessment, and multimodal interview analysis while maintaining human oversight and fairness controls.

References

[1] Elviwani, Siahaan, A. P. U., & Fitriana, L. (n.d.). *Performance-based Stable Matching using Gale–Shapley Algorithm*.

[2] Suakanto, S., Siswanto, J., Prasetyo, I. R., Hardiyanti, M. (2021). *Interview Bot for Improving Human Resource Management*. In 2021 Int. Conf. on ICT for Smart Society (ICISS), pp. 4644–4647. IEEE. DOI: 10.1109/ICISS53185.2021.9533248.

[3] Mendoza, A. A. D., Barbosa, L. J. C., Cortez, D. M. A., Dioses, R. M., Agustin, V. A., Regala, R. C. (2022). *Enhancement of Gale–Shapley algorithm with imbalanced sets for hiring and job finding applications*. Indonesian Journal of Electrical Engineering and Computer Science, 27(2), 954–958. DOI: 10.11591/ijeecs.v27.i2.pp954-958.

[4] Will, P., Krpan, D., & Lordan, G. (2023). *People versus Machines: Introducing the HIRE Framework*. Artificial Intelligence Review, 56, 1071–1100. Accepted: 13 April 2022. DOI: 10.1007/s10462-022-10193-6.

[5] Thangaramya, K., Roselind, J. D., Logeswari, G., Ahirwar, N., & Gajendran, S. (2024). *Automated Resume Parsing and Ranking using Natural Language Processing*. In 2024 3rd IEEE Int. Conf. on Artificial Intelligence for Internet of Things (AIIoT). DOI: 10.1109/AIIoT58432.2024.10574696.

[6] Hosseini, H., Roy, S., & Zhang, D. (2024). *Putting Gale & Shapley to Work: Guaranteeing Stability Through Learning*. In 38th Conf. on Neural Information Processing Systems (NeurIPS 2024).

[7] Aly, A. M., El Alem, A., Alawida, M., AlBloushi, S., & Aldhaen, G. (2025). *Optimizing Recruitment with AI: A Smarter Approach to Candidate Evaluation*. In

2025 1st Int. Conf. on Computational Intelligence Approaches and Applications (ICCIAA). DOI: 10.1109/ICCIAA65327.2025.11013684.

[8] Mujtaba, D. F., & Mahapatra, N. R. (2025). *Fairness in AI-Driven Recruitment: Challenges, Metrics, Methods, and Future Directions.* arXiv preprint arXiv:2405.19699v3.

Chapter A

Appendix A: Example prompts and rubric

A.1 LLM prompt templates

A.2 Scoring rubric